

SynthDrive v0.1: Formal Specification

Synthetic Telematics Generatorfor Motor Insurance

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Abstract

SynthDrive v0.1 is a synthetic personal auto telematics data generator. This document gives a complete algebraic description of its sampling pipeline, comprising a Gaussian copula for joint feature generation, a zero-inflated negative binomial model for claim frequency, a Gamma model for aggregate claim severity, and a deterministic constraint-enforcement pass. The generator is calibrated to the structure and statistics of the public So-Boucher-Valdez (2021) synthetic telematics dataset (100,000 policies), and produces output portfolios with 55 columns. All modeling assumptions, calibrated parameters, and constraint rules are stated explicitly so that the specification can serve as the authoritative reference for the SynthDrive codebase.

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1 Introduction

This document is the formal algebraic specification for SynthDrive v0.1, an open-source synthetic personal auto telematics data generator. It is intended to serve as the authoritative reference for the SynthDrive codebase and as a companion to the methodological working paper (Homayounfar, 2026).

The specification describes the complete sampling pipeline in precise mathematical notation: the Gaussian copula for joint feature generation (Section 3), categorical variable sampling (Section 4), duration and exposure (Section 5), derived telematics variables (Section 6), claim frequency via a zero-inflated negative binomial model (Section 7), claim severity via a Gamma model (Section 8), pure premium (Section 9), and the deterministic constraint-enforcement pass (Section 10). The two operating modes—seed mode and parameter mode—are described in Section 11.

All calibrated parameters reflect the structure and statistics of the public So–Boucher–Valdez (So et al., 2021) synthetic telematics dataset.

SynthDrive output is twice removed from real insurer data: the public So–Boucher–Valdez dataset was itself generated from a proprietary Canadian insurer portfolio, and SynthDrive is in turn calibrated from that synthetic dataset. SynthDrive therefore makes no claim to reproduce real driving behavior or real insurer loss statistics. Full data provenance is documented in the companion paper.

2 Notation and Index Set

Let $\mathcal{I} = \{1, \dots, n\}$ be the portfolio index set, where n is the requested portfolio size.

For each policy $i \in \mathcal{I}$, the data object is the tuple

$$\left(X_i^c, X_i^d, e_i, T_i, N_i, A_i, \text{PP}_i \right)$$

where:

- $X_i^c \in \mathbb{R}^9$ is the continuous feature vector (Section 3),
- X_i^d is the discrete/categorical feature vector (Section 4),
- $e_i \in (0, 1]$ is the exposure in policy-years (Section 5),
- T_i is the derived telematics feature vector (Section 6),
- $N_i \in \mathbb{Z}_{\geq 0}$ is the claim count (Section 7),
- $A_i \in \mathbb{R}_{\geq 0}$ is the aggregate claim amount (Section 8),
- $\text{PP}_i \in \mathbb{R}_{\geq 0}$ is the pure premium (Section 9).

All random variables are defined on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The generator is seeded by a single integer `random_state`, which initialises numpy's `default_rng`.

3 Continuous Feature Distribution: Gaussian Copula

The continuous feature vector X_i^c captures the nine variables that drive both the portfolio structure and the claim models. Their joint distribution is modeled by a Gaussian copula, which separates the marginal distribution of each variable from the rank correlation structure of the joint distribution. This approach is fully interpretable: every modeling assumption reduces to a marginal parameter or a pairwise Spearman correlation.

3.1 Copula variables

The continuous copula vector for policy i is

$$X_i^c = \left(\text{age}_i, \text{credit}_i, \text{yncl}_i, \text{cage}_i, \text{miles}_i, \text{days}_i, \text{pct}_i, \text{accel}_{9,i}, \text{brake}_{9,i} \right)^\top \in \mathbb{R}^9.$$

Here $\text{accel}_{9,i}$ and $\text{brake}_{9,i}$ are the raw event counts at the 9 mph/s threshold, serving as anchor variables from which the full threshold sequences are derived (Section 6.4).

The joint distribution is constructed via a Gaussian copula.

3.2 Spearman-to-Pearson conversion

Let $\Sigma^S \in [-1, 1]^{9 \times 9}$ be the target Spearman rank correlation matrix (specified in `presets/core.py` and, in seed mode, estimated from the SBV data). The Pearson correlation matrix used by the Gaussian copula is obtained element-wise by

$$\rho_{jk}^P = 2 \sin \left(\frac{\pi \rho_{jk}^S}{6} \right), \quad j, k = 1, \dots, 9, \quad (1)$$

with $\rho_{jj}^P = 1$.

If the resulting matrix Σ^P is not positive definite, it is replaced by the nearest positive-definite correlation matrix via the Higham (2002) procedure (eigenvalue flooring at $\epsilon = 10^{-8}$ followed by diagonal rescaling).

3.3 Sampling

Let $\Sigma^P = LL^\top$ be the Cholesky factorisation. For each policy i :

$$Z_i \sim \mathcal{N}(0, I_9), \quad (2)$$

$$\tilde{Z}_i = LZ_i, \quad (3)$$

$$U_{ij} = \Phi(\tilde{Z}_{ij}), \quad j = 1, \dots, 9, \quad (4)$$

where Φ is the standard normal CDF. The vector $U_i = (U_{i1}, \dots, U_{i9})^\top$ has uniform marginals on $(0, 1)$ and rank correlations that approximate Σ^S .

3.4 Marginal inverse CDFs

Each component $X_{ij}^c = F_j^{-1}(U_{ij})$, where F_j is the marginal CDF for variable j . The parametric marginals in the `core` preset are given in Table 1.

Table 1: Marginal distributions for continuous copula variables (parameter mode). Car age uses a shifted distribution to accommodate the observed range $[-2, 20]$. Accel and brake anchors are raw event `counts`.

Variable	Distribution	Key parameters	Support
age	Truncated normal	$\mu = 45, \sigma = 14$	$[16, 90]$
credit	Truncated normal	$\mu = 650, \sigma = 85$	$[400, 900]$
yncl	Exponential	mean = 4	$[0, 50]$
cage	Shifted gamma	shape = 2.5, scale = 3.2, shift = -2	$[-2, 20]$
miles	Log-normal	$\mu_{\log} = 9.35, \sigma_{\log} = 0.42$	$[500, 60\,000]$
days	Truncated normal	$\mu = 4.6, \sigma = 1.2$	$[0, 7]$
pct	Beta	$\alpha = 2.2, \beta = 4.5$	$(0, 1)$
accel ₉	Log-normal	$\mu_{\log} = 3.18, \sigma_{\log} = 1.20$	$\mathbb{R}_{\geq 0}$, integer
brake ₉	Log-normal	$\mu_{\log} = 1.79, \sigma_{\log} = 0.85$	$\mathbb{R}_{\geq 0}$, integer

Integer variables (age, credit, yncl, cage, accel₉, brake₉) are rounded and clipped to their stated bounds after transformation.

In seed mode, F_j^{-1} is replaced by an empirical inverse CDF estimated from the supplied seed CSV.

4 Categorical Variables

The discrete feature vector $X_i^d = (\text{sex}_i, \text{marital}_i, \text{use}_i, \text{terr}_i, \text{reg}_i)$ is sampled independently of X_i^c . This is a simplifying assumption reflecting the weak rank correlations observed between categorical and continuous variables in the SBV data. All categorical values are stored as strings matching the

SBV column values exactly. Empirical proportions are taken from the SBV dataset ($n = 100\,000$).

$$\begin{aligned} \text{sex}_i &\sim \text{Categorical}(p_{\text{Male}} = 0.539, p_{\text{Female}} = 0.461), \\ \text{marital}_i &\sim \text{Categorical}(p_{\text{Married}} = 0.699, p_{\text{Single}} = 0.301), \\ \text{use}_i &\sim \text{Categorical}(p_{\text{Commute}} = 0.498, p_{\text{Private}} = 0.461, p_{\text{Commercial}} = 0.026, p_{\text{Farmer}} = 0.014), \\ \text{reg}_i &\sim \text{Categorical}(p_{\text{Urban}} = 0.781, p_{\text{Rural}} = 0.219). \end{aligned}$$

Territory terr_i is an integer-valued zone code with observed range $[11, 91]$, mean 56.5, and standard deviation 24.0. In parameter mode it is sampled from an empirical discrete distribution estimated from the SBV data. It is not a six-level categorical variable.

5 Duration and Exposure

In the SBV dataset, **Duration** is recorded in days (integer values). Exposure in policy-years is derived as:

$$e_i = \frac{\text{Duration}_i}{365}, \quad e_i \in (0, 1]. \quad (5)$$

In parameter mode, duration is drawn from a discrete mixture representing common policy terms expressed in days:

$$\text{Duration}_i \sim 0.60 \cdot \delta_{365} + 0.30 \cdot \delta_{182} + 0.10 \cdot \delta_{91},$$

where δ_v denotes a point mass at v days. Exposure is then computed from (5).

6 Derived Telematics Variables

Several telematics variables are not sampled directly but are derived deterministically from X_i^c and e_i after the copula draw. These include total miles driven, day-of-week driving proportions, time-of-day proportions, hard-event threshold sequences, and turn intensity sequences. Deriving rather than sampling these variables ensures that internal consistency constraints—compositional proportions, monotone threshold sequences, and mileage coherence—are satisfied by construction rather than by post-hoc correction.

6.1 Total miles driven

$$\text{total_miles}_i = \text{miles}_i \cdot e_i \cdot (0.6 + 0.8 \cdot \text{pct}_i) \cdot \exp(\varepsilon_i^m), \quad \varepsilon_i^m \sim \mathcal{N}(0, 0.12^2). \quad (6)$$

6.2 Day-of-week composition

Let $\alpha^{\text{dow}} \in \mathbb{R}_{>0}^7$ be a fixed concentration vector (weekdays weighted slightly higher than weekends). The daily driving proportions are:

$$(p_{\text{mon},i}, \dots, p_{\text{sun},i})^\top \sim \text{Dirichlet}(\alpha^{\text{dow}}), \quad (7)$$

so that $\sum_d p_{d,i} = 1$ by construction. Column names follow the SBV convention: Thursday is **Pct.drive.thr**.

Derived aggregates:

$$p_{\text{wkday},i} = \sum_{d \in \{\text{mon}, \text{tue}, \text{wed}, \text{thr}, \text{fri}\}} p_{d,i}, \quad p_{\text{wkend},i} = p_{\text{sat},i} + p_{\text{sun},i}.$$

All day-of-week proportions are stored as fractions in $[0, 1]$.

6.3 Time-of-day variables

Five time-of-day proportion variables are sampled from independent Beta distributions. Column names follow the SBV convention exactly, including the space in the rush-hour names:

$$\begin{aligned} p_{\text{rush am},i} &\sim \text{Beta}(3.0, 12.0), \\ p_{\text{rush pm},i} &\sim \text{Beta}(4.2, 10.0), \\ p_{2\text{hrs},i} &\sim \text{Beta}(0.6, 12.0), \\ p_{3\text{hrs},i} &\sim \text{Beta}(0.5, 14.0), \\ p_{4\text{hrs},i} &\sim \text{Beta}(0.5, 14.0). \end{aligned}$$

These represent fractions of total driving in specific time windows and are stored as proportions in $[0, 1]$. They are not constrained to sum to any particular value.

6.4 Hard-event threshold sequences

Unit of measurement. Accel and brake columns are *raw event counts* (non-negative integers), not rates per mile. This matches the SBV dataset, where for example `Accel.06miles` has mean 43 and maximum 621.

Threshold set. The threshold set is $\mathcal{K} = \{6, 8, 9, 11, 12, 14\}$ mph/s. There is no 10 mph/s column. The top threshold is 14 mph/s.

Sampling. The copula supplies anchor counts $\text{accel}_{9,i}$ and $\text{brake}_{9,i}$ at the 9 mph/s threshold. For each $k \in \mathcal{K}$, let r_k^{acc} (respectively r_k^{brk}) be the pre-calibrated ratio for threshold k relative to the 9 mph/s anchor. Then:

$$\tilde{a}_{k,i} = \text{accel}_{9,i} \cdot r_k^{\text{acc}} \cdot \exp(\varepsilon_{k,i}), \quad \varepsilon_{k,i} \sim \mathcal{N}(0, 0.15^2), \quad (8)$$

rounded to the nearest non-negative integer. The enforced (non-increasing) sequence is then:

$$\text{accel}_{k,i} = \text{cummin}_{k \in \mathcal{K}}(\tilde{a}_{k,i}), \quad (9)$$

i.e. the cumulative minimum taken from $k = 6$ to $k = 14$. The same construction applies to $\text{brake}_{k,i}$.

6.5 Turn intensity sequences

Turn intensity columns are also raw event counts. The SBV dataset shows `Left.turn.intensity08` with mean 916 and maximum 794 740, and `Right.turn.intensity08` with mean 2 006 and maximum 841 144.

Left-turn base count $\text{lt}_{08,i}$ and right-turn base count $\text{rt}_{08,i}$ are sampled jointly via a bivariate Gaussian copula with Spearman correlation ρ_{lr}^S (estimated from the seed data), converted to Pearson by (1). Log-normal marginals are calibrated from the seed data.

Sub-threshold counts at $g \in \{0.8, 0.9, 1.0, 1.1, 1.2\}$ g are derived analogously to (8)–(9) using step ratios and noise calibrated from the seed data, then rounded to non-negative integers.

7 Claim Frequency: Zero-Inflated Negative Binomial

Claim count N_i is generated in Stage 2 of the pipeline, conditional on the full feature vector (X_i^c, X_i^d, e_i) produced in Stage 1. The zero-inflated negative binomial (ZINB) model is used because motor insurance claim counts exhibit two distinct sources of zeros: structural zeros from low-risk policies that would never claim under the modeled conditions, and sampling zeros from policies that simply did not claim during the observation period. The ZINB model handles both through separate structural-zero and count components.

7.1 Model structure

The claim count N_i follows a zero-inflated negative binomial (ZINB) model:

$$N_i = S_i \cdot 0 + (1 - S_i) \cdot \tilde{N}_i, \quad (10)$$

where $S_i \sim \text{Bernoulli}(\pi_i)$ is the structural-zero indicator (independent of $\tilde{N}_i$) and $\tilde{N}_i \sim \text{NegBin}(\mu_i, \alpha)$.

7.2 Zero-inflation probability

$$\text{logit } \pi_i = \gamma_0 + \gamma_y \cdot \text{yncl}_i + \gamma_c \cdot \frac{\text{credit}_i - 650}{100}, \quad (11)$$

with calibrated parameters $\gamma_0 = -1.735$, $\gamma_y = 0.035$, $\gamma_c = 0.12$. The value $\text{expit}(-1.735) \approx 0.15$ gives the structural-zero probability for the reference driver ($\text{yncl} = 0$, $\text{credit} = 650$).

7.3 Negative binomial mean

The NB mean uses a log link with an exposure offset:

$$\log \mu_i = \log e_i + \log \lambda_0 + f(X_i), \quad (12)$$

where $\lambda_0 = 0.0165$ is the base annual claim rate and $f(X_i)$ is the additive log-relative-risk score:

$$\begin{aligned} f(X_i) = & \beta_{\text{yng}} \cdot \mathbf{1}[\text{age}_i < 25] + \beta_{\text{sen}} \cdot \mathbf{1}[\text{age}_i > 65] + \beta_{\text{male}} \cdot \mathbf{1}[\text{sex}_i = \text{Male}] \\ & + \beta_{\text{snl}} \cdot \mathbf{1}[\text{marital}_i = \text{Single}] + \beta_{\text{cred}} \cdot \frac{\text{credit}_i - 650}{100} \\ & + \beta_{\text{com}} \cdot \mathbf{1}[\text{use}_i = \text{Commercial}] + \beta_{\text{far}} \cdot \mathbf{1}[\text{use}_i = \text{Farmer}] \\ & + \beta_{\text{mi}} \cdot (\log \text{miles}_i - \log 12\,000) + \beta_{\text{pct}} \cdot (\text{pct}_i - 0.33) \\ & + \beta_{\text{cage}} \cdot (\text{cage}_i - 9) \\ & + \beta_{\text{brk}} \cdot (\log(1 + \text{brake}_{9,i}) - \log(1 + \tilde{b}_9)) + \beta_{\text{acc}} \cdot (\log(1 + \text{accel}_{9,i}) - \log(1 + \tilde{a}_9)) + \delta(\text{terr}_i). \end{aligned} \quad (13)$$

Here $\tilde{b}_9$ and $\tilde{a}_9$ are the empirical medians of brake_9 and accel_9 in the SBV dataset ($\tilde{b}_9 = 2$, $\tilde{a}_9 = 0$). The $\log(1 + \cdot)$ transformation is applied before centering to prevent extreme raw count outliers (observed up to 800+) from producing unbounded risk scores. The territory effect $\delta(\text{terr}_i)$ is a smooth function of the numeric territory code, e.g. a linear or step-function map calibrated from the data.

The calibrated coefficients are given in Table 2.

Table 2: Frequency model log-relative-risk coefficients. Median centering constants $\tilde{b}_9 = 2$ and $\tilde{a}_9 = 0$ are estimated from the SBV data.

Symbol	Description	Value
λ_0	Base annual rate	0.0165
α	NB overdispersion	0.30
β_{yng}	Young driver (age < 25)	-0.28
β_{sen}	Senior driver (age > 65)	+0.40
β_{male}	Male	+0.04
β_{snl}	Single	+0.10
β_{cred}	Per 100 credit pts above 650	-0.28
β_{com}	Commercial vehicle use	+0.35
β_{far}	Farmer vehicle use	-0.53
β_{mi}	Log annual miles (per log-unit)	+0.22
β_{pct}	Annual pct driven (per unit above 0.33)	+2.40
β_{cage}	Vehicle age (years above 9)	-0.07
β_{brk}	Hard braking, $\log(1 + \text{brake}_9)$ centred at $\log(1 + \tilde{b}_9)$	+0.25
β_{acc}	Hard acceleration, $\log(1 + \text{accel}_9)$ centred at $\log(1 + \tilde{a}_9)$	+0.18
$\delta(\cdot)$	Territory log-RR (numeric zone code)	calibrated

7.4 NB variance and sampling

The NB parameterisation satisfies

$$\text{Var}[\tilde{N}_i] = \mu_i + \alpha\mu_i^2.$$

Because $\alpha = 0.30$ is not an integer, sampling uses the exact gamma–Poisson mixture representation:

$$\tilde{\lambda}_i \sim \text{Gamma}\left(\frac{1}{\alpha}, \alpha\mu_i\right), \quad \tilde{N}_i \sim \text{Poisson}(\tilde{\lambda}_i). \quad (14)$$

This is exact for any $\alpha > 0$, including non-integer values.

7.5 Marginal zero probability

The marginal probability of zero claims is

$$\mathbb{P}(N_i = 0) = \pi_i + (1 - \pi_i) \mathbb{P}(\tilde{N}_i = 0) = \pi_i + (1 - \pi_i) \left(\frac{1}{1 + \alpha\mu_i} \right)^{1/\alpha}.$$

8 Claim Severity: Gamma Model

Aggregate claim amount A_i is generated in Stage 3, conditional on $N_i \geq 1$. Severity is modeled separately from frequency, following the standard actuarial frequency-severity decomposition. The Gamma distribution is chosen for its support on $(0, \infty)$, its right-skewed shape, and its tractability; the shape parameter scales linearly with claim count so that multi-claim policies have proportionally larger expected amounts.

8.1 Model structure

Aggregate severity is defined conditionally on having at least one claim. For $N_i = n \geq 1$:

$$A_i \mid N_i = n, X_i \sim \text{Gamma}(\kappa \cdot n, \theta \cdot g_i), \quad (15)$$

where $\kappa = 2$ is the shape parameter per claim and $\theta = 1567$ is the base scale parameter. The shape increases linearly with claim count, so $\mathbb{E}[A_i \mid N_i = n, X_i] = \kappa n \theta g_i$.

For the reference driver ($g_i = 1$) with $n = 1$ claim:

$$\mathbb{E}[A_i \mid N_i = 1, \text{reference}] = \kappa \theta = 3134.$$

For $N_i = 0$: $A_i = 0$ unconditionally.

8.2 Risk-adjusted scale factor

$$\log g_i = \beta_{\text{yng}}^A \cdot \mathbf{1}[\text{age}_i < 25] + \beta_{\text{mi}}^A \cdot (\log \text{miles}_i - \log 12000) + \beta_{\text{ca}}^A \cdot (\text{cage}_i - 7), \quad (16)$$

with $\beta_{\text{yng}}^A = 0.20$, $\beta_{\text{mi}}^A = 0.12$, $\beta_{\text{ca}}^A = 0.018$.

Note that the centering value for car age is 7 years, which lies within the observed range $[-2, 20]$ and is close to the empirical mean of 5.6 years in the SBV data.

8.3 Minimum threshold

A hard floor is applied post-sampling:

$$A_i \leftarrow \max(A_i, 1) \quad \text{for } N_i \geq 1.$$

9 Pure Premium

The pure premium (annualised expected loss cost per unit of exposure) is

$$\text{PP}_i = \frac{A_i}{e_i}. \quad (17)$$

Because $e_i > 0$ by construction (Section 5), (17) is always defined. When $N_i = 0$ we have $A_i = 0$, hence $\text{PP}_i = 0$.

10 Constraints

After all sampling is complete, the following constraints are enforced deterministically and in the order listed.

10.1 Scalar bounds

$$\begin{aligned}
e_i &\in (0, 1], \\
\text{age}_i &\in [16, 90], \\
\text{credit}_i &\in [400, 900], \\
\text{yncl}_i &\geq 0, \\
\text{cage}_i &\in [-2, 20], \quad (\text{negative values reflect model-year convention}), \\
\text{miles}_i &\in [500, 60\,000], \\
\text{pct}_i, p_{d,i}, p_{\text{weekday},i}, p_{\text{wkend},i}, p_{\text{rush},i} &\in [0, 1], \\
\text{total_miles}_i &\geq 0, \\
\text{accel}_{k,i}, \text{brake}_{k,i}, \text{lt}_{g,i}, \text{rt}_{g,i} &\in \mathbb{Z}_{\geq 0}. \quad (\text{non-negative integers}).
\end{aligned}$$

10.2 Compositional constraint

$$\sum_{d \in \{\text{mon}, \text{tue}, \text{wed}, \text{thr}, \text{fri}, \text{sat}, \text{sun}\}} p_{d,i} = 1. \tag{18}$$

If any row sum is zero, the vector is replaced by the uniform distribution $p_{d,i} = 1/7$. After normalisation, $p_{\text{weekday},i}$ and $p_{\text{wkend},i}$ are recomputed as defined in Section 6.

10.3 Monotone threshold sequences

For each policy i , the accel sequence must be non-increasing across the threshold set $\mathcal{K} = \{6, 8, 9, 11, 12, 14\}$:

$$\text{accel}_{6,i} \geq \text{accel}_{8,i} \geq \text{accel}_{9,i} \geq \text{accel}_{11,i} \geq \text{accel}_{12,i} \geq \text{accel}_{14,i} \geq 0. \tag{19}$$

Note: there is no 10 mph/s column. The same holds for the brake sequence. Turn intensity sequences are non-increasing across $\{0.8, 0.9, 1.0, 1.1, 1.2\}$ g. Enforcement uses the cumulative minimum.

10.4 Claim consistency

$$N_i \in \mathbb{Z}_{\geq 0} \quad (\text{non-negative integer}), \tag{20}$$

$$A_i \geq 0, \tag{21}$$

$$A_i = 0 \iff N_i = 0. \tag{22}$$

10.5 Pure premium recomputation

After enforcing (22), PP_i is recomputed from (17).

11 Seed Mode vs. Parameter Mode

SynthDrive v0.1 operates in one of two modes.

Parameter mode. All parameters ($F_j^{-1}, \Sigma^S, \lambda_0, \alpha, \dots$) are taken from the calibrated defaults in `synthdrive/presets/core.py`. No external data are required. This mode is the fallback when the seed CSV is absent or unreadable.

Seed mode. If the path to `telematics_syn-032021.csv` is supplied via `seed_path`, the Spearman matrix Σ^S and the marginal CDFs F_j are estimated from the seed data via empirical rank correlations and empirical quantile functions, respectively. Categorical proportions and the territory distribution are also taken from the seed data. All frequency and severity model coefficients remain at their parameter-mode defaults unless explicitly overridden.

If the seed file is absent or unreadable, the generator falls back to parameter mode and issues a `UserWarning`.

12 Summary of Parameter Symbols

Table 3: Complete list of free parameters in SynthDrive v0.1.

Symbol	Description	Section
$\Sigma^S \in [-1, 1]^{9 \times 9}$	Spearman correlation matrix	3
$F_j^{-1}, j = 1, \dots, 9$	Marginal inverse CDFs	3
$(p_{\text{use},k}), (p_{\text{sex},k}), \text{etc.}$	Categorical proportions	4
(v_k, w_k)	Duration mixture values and weights	5
$\alpha^{\text{dow}} \in \mathbb{R}_{>0}^7$	Dirichlet concentration vector	6
$r_k^{\text{acc}}, r_k^{\text{brk}}, k \in \mathcal{K}$	Accel/brake threshold ratios	6
$\gamma_0, \gamma_y, \gamma_c$	Zero-inflation logistic coefficients	7
λ_0	Base annual claim rate	7
α	NB overdispersion parameter	7
$\beta_{\text{yng}}, \dots, \delta(\cdot)$	Frequency log-RR coefficients	7
κ, θ	Severity shape and base scale	8
$\beta_{\text{yng}}^A, \beta_{\text{mi}}^A, \beta_{\text{ca}}^A$	Severity scale-factor coefficients	8

References

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